RONALD EPHRAIM C. TIANGSON

Senior Software Engineer | Full-Stack Engineering | Database Modernisation | Cloud | AI Systems

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PROFESSIONAL SUMMARY

Senior Software Engineer with 16+ years of experience delivering and modernising enterprise software across banking, finance, insurance, healthcare, SaaS, and knowledge-management environments. Strong hands-on background in C#/.NET, Python/FastAPI, React, Angular, REST APIs, SQL and database engineering, database migration, cloud/DevOps, automated testing, reporting, and AI/LLM-enabled systems. Experienced across the full SDLC - requirements, architecture, implementation, testing, deployment, production support, and continuous improvement - with additional experience mentoring developers, reviewing code, and working directly with technical and business stakeholders. Proven record of modernising legacy systems and improving data-heavy workloads, including reducing a long-running batch process from 23 hours to 2-3 hours and an insurance reporting process from 2-3 months to 1 day. Completed a Master of Software Engineering (Level 9) in New Zealand in August 2026.

TECHNICAL SKILLS

Programming Languages & Legacy: C#, .NET Core, .NET Framework, ASP.NET, Classic ASP, Python, TypeScript, JavaScript, VB.NET, VBA, PHP, SQL; legacy framework support and application modernisation

Frontend & UI: React, Angular, HTML5, CSS3, Tailwind CSS, Bootstrap, Kendo UI, Vite, responsive UI development, reusable component design

Backend, APIs & Architecture: FastAPI, ASP.NET Web API, REST APIs, API documentation, Postman, Microservices, Clean Architecture, Layered Architecture, SOA, DTO/Mapper/Repository patterns, Service Layer, Dependency Injection, SOLID principles, OOP, MVC, MVVM

Databases, Migration & Reporting: SQL Server, PostgreSQL, Supabase, MySQL, SQLite, Oracle, DB2, pgvector, SQL query tuning, stored procedures, indexing, database architecture, database migration, SQLAlchemy, Entity Framework, SSIS, SSRS, Crystal Reports

AI & Data Engineering: LLM integration, Agentic AI systems, autonomous AI agents, agentic automation loops (plan-act-observe-refine), AI workflow orchestration, tool/function calling, AI connectors (Gmail, OAuth 2.0, REST APIs, external knowledge sources), Retrieval-Augmented Generation (RAG), graph-enhanced RAG, vector databases (FAISS, pgvector), semantic search, embeddings, prompt engineering, multi-model AI routing, context-aware AI memory, OCR/document parsing, AI-driven task automation, data analysis, and exploratory data analysis (EDA).

AI Development Tools & Assistants: OpenAI Codex, ChatGPT, Claude, Hermes, Sorcery (AI-powered code review and refactoring), AI-assisted coding, agentic coding workflows, automated code review, code quality analysis, code generation, debugging, refactoring, automated testing, and AI-assisted software development.

Cloud & DevOps: AWS Lambda, API Gateway, RDS, Oracle Cloud Free Tier, Microsoft Azure, Azure Static Web Apps, Docker, Kubernetes, CI/CD, GitHub Actions, Azure DevOps, Bamboo, Git, GitHub, environment-based configuration

Testing & Quality: xUnit, NUnit, SpecFlow, Selenium, Cypress, PyTest, Moq, unit testing, integration testing, regression testing, UI automation, code review, secure development practices

Security & Application Practices: Authentication and authorization, JWT-based security, bcrypt password hashing, role-aware access control, input validation, secure data handling, privacy-aware design, error handling

Leadership & Delivery: Requirements analysis, technical planning, Agile/Scrum, Kanban, mentoring, junior developer leadership, stakeholder collaboration, technical documentation, source-control workflow, sprint planning, production support

Legacy Modernisation & Integration: Classic ASP and legacy .NET support, ASP.NET modernisation, Mainframe/DB2-to-SQL Server migration, monolith-to-microservices conversion, REST/API integration, legacy reporting modernisation, batch-processing optimisation

PROFESSIONAL EXPERIENCE

Software Engineering Intern (Volunteer)

Tiaki Taonga Trust Wai 262 | Auckland, New Zealand | Apr 2026 - Aug 2026 | 15-week internship

Contributed to a professional portfolio platform supporting structured digital records, secure access, stakeholder workflows, and long-term information management for the Trust.

- Supported junior developers by assigning work, reviewing code, explaining design decisions, and improving clean-code, debugging, testing, and delivery practices.
- Planned development work across requirements analysis, feature breakdown, sprint priorities, technical tasks, risk identification, and delivery sequencing.
- Created and refined functional requirements, user stories, acceptance criteria, workflow definitions, and implementation notes from stakeholder discussions.
- Designed and developed portfolio-management capabilities using maintainable full-stack architecture for future extension and operational use.
- Built features across the React/TypeScript interface, FastAPI backend, APIs, data models, validation rules, and integration points.
- Applied secure-by-design practices for authentication, authorization, input validation, data privacy, error handling, and controlled access to sensitive information.

- Established standards for reusable components, naming, API contracts, documentation, source-control workflow, review-ready pull requests, testing, refactoring, and deployment readiness.

Application Analyst Developer

Satellite Office - Smartgroup Australia | Manila, Philippines | Jul 2023 - Jul 2025

Maintained and enhanced enterprise salary-packaging and financial platforms supporting large-scale Australian clients across corporate, healthcare, government, and education sectors.

- Optimised stored procedures and batch processing, reducing a long-running data process from 23 hours to 2-3 hours through indexing, query tuning, and batch refactoring.
- Developed and enhanced SSRS and PDF Pro reports, providing operational visibility and data insights to business units.
- Led feature development and system enhancements that introduced new business functionality and improved user experience.
- Troubleshot application and data issues, resolving incidents efficiently to minimise downtime and maintain system stability.
- Provided technical support to internal and external clients, including tickets, user issues, and change requests.
- Participated in code reviews, coding standards, unit tests, integration tests, and automated test suites for new and legacy applications.
- Worked with project managers, business analysts, and end users to translate business requirements into reliable technical solutions.

Senior Software Engineer

Willis Towers Watson | Manila, Philippines | Mar 2020 - Jul 2023

Developed and supported enterprise HR and benefits platforms including BrightChoices, Benefit Access, and FSA Portal for global employee insurance and wellness programmes.

- Led feature development and enhancement work using .NET Core, Angular, SQL Server, REST APIs, and Azure DevOps.
- Participated in extracting and converting monolithic components into microservices, improving modularity, deployment speed, and maintainability.
- Designed and implemented APIs, data modules, and backend services integrated with existing enterprise platforms.
- Applied SOLID principles, repository and service layers, factory patterns, DTOs, dependency injection, and modular architecture practices.
- Participated in code reviews, maintained coding standards, and promoted engineering consistency across development teams.
- Mentored and coached junior developers in clean code, debugging, version control, testing, and Agile delivery.
- Participated in technical hiring by assessing candidate skills, conducting interviews, and supporting hiring decisions.
- Delivered unit, integration, and UI automation tests using NUnit, SpecFlow, Selenium, and Cypress.
- Hosted knowledge-sharing sessions covering frameworks, tools, and architecture practices.

Core Software Developer

AXA Philippines | Manila, Philippines | Aug 2018 - Mar 2020

Worked on ACE-DNN Solutions Delivery, a backend insurance automation platform focused on reducing manual processing and producing faster, more accurate reporting.

- Developed and integrated .NET Core modules, improving system performance and user experience.
- Automated manual reporting and data-generation workflows, reducing a 2-3 month operational process to 1 day.
- Refactored legacy SQL and optimised database queries, improving average processing performance by approximately 45%.
- Participated in Change Advisory Board reviews, presenting planned changes, risks, impacts, and deployment requirements.
- Handled on-premises deployments and environment configuration for production rollouts.
- Conducted code reviews and maintenance activities to keep applications stable, secure, and aligned with best practices.
- Worked with QA engineers and business analysts to resolve issues early and improve release quality; maintained unit and integration test coverage.

Full Stack Developer

L.R.B. Medical & Maternity Clinic | Manila, Philippines | Jan 2018 - Aug 2018

Built and maintained a clinic-management system supporting patient records, billing, scheduling, and reporting.

- Developed full-stack workflows using C#, .NET Core, Angular, SQL Server, and Entity Framework.
- Optimised data flow and PSA birth-record submissions, reducing manual reporting time by more than 50%.
- Implemented secure patient-data storage and access layers for real-time clinical and administrative use.
- Managed the SDLC from requirements gathering and architecture design through testing, deployment, and staff onboarding.

C#/.NET Developer

Eclaro International - OneBeacon Insurance Group | Manila, Philippines | May 2015 - Dec 2017

Supported and modernised enterprise applications for OneBeacon Insurance Group, a US-based property and casualty insurance organisation.

- Modernised legacy ASP.NET systems by upgrading front-end components from HTML4 to HTML5 and integrating TypeScript and Angular.
- Developed and integrated RESTful APIs to improve front-end and back-end communication.
- Led database migration from Mainframe/DB2 to SQL Server, optimising stored procedures, SQL scripts, and indexes.
- Implemented automated testing frameworks using xUnit and NUnit and led Selenium test development.
- Managed CI/CD pipelines using Atlassian Bamboo.
- Enhanced and maintained more than 100 SSRS reports, improving reporting accuracy and executive visibility.

- Established and enforced .NET coding standards for maintainability, scalability, and team consistency.
- Re-engineered core business modules and reporting pipelines, achieving up to 35% faster system performance.

Lead Software Analyst

Equitable Computer Services Inc. | Manila, Philippines | Mar 2008 - May 2015

Developed enterprise banking and financial applications supporting major Philippine institutions including Equitable Bank, Robinsons Bank, and BDO.

- Led a team of three junior developers, mentoring them in coding standards, debugging, requirements gathering, deployment, and full system development.
- Designed and developed enterprise banking applications and supporting data processes for Equitable Bank, Robinsons Bank, and BDO.
- Developed the Credit Card Management System from the ground up, including database architecture, workflow automation, reporting modules, and backend processing.
- Owned backend data processes generating financial, reconciliation, compliance, credit-card, and debit-card reports.
- Collaborated directly with senior executives to design and implement client-facing reward and raffle systems for banks and insurance partners.
- Built systems for sister companies including myHealth and Medilink, integrating healthcare-management and billing workflows with financial platforms.
- Delivered high-volume ASP.NET and Windows applications with focus on data security, scalability, operational reliability, and long-term maintainability.
- Worked across legacy technologies and modernisation activities during the early part of the career, including Classic ASP, VBA/VB-based tooling, .NET Framework, SQL Server, and reporting/database workflows.

SELECTED PROJECTS

MemoLink - Context-Aware AI Companion for Knowledge Capture, Retrieval & Task Support

Master of Software Engineering capstone. Full-stack knowledge platform using React, TypeScript, FastAPI, SQLAlchemy, PostgreSQL/Supabase, pgvector, REST APIs, and multi-model LLM integration. Includes RAG with source citations, persistent chat history, file ingestion, embeddings/vector search, AI-assisted note and task workflows, translation, web/academic search integrations, feature flags, admin tools, and a OneDrive-backed Books Library. Uses Clean Architecture, repository/service patterns, dependency injection, validation, JWT/bcrypt security, and deployment-ready configuration.

Wai 262 Portfolio Management System

Professional portfolio platform for Tiaki Taonga Trust Wai 262 supporting structured digital records, project artefacts, stakeholder information, role-aware access, and secure knowledge workflows. Tech: React, TypeScript, Python, FastAPI, REST APIs, SQLAlchemy, Azure, GitHub, CI/CD, and Agile delivery. Practices included requirements analysis, sprint planning, secure data handling, code review, mentoring, and maintainable documentation.

KaiHelper - AI Grocery Budgeting App

AI-powered budgeting application developed during the Master of Software Engineering. Parses grocery receipts, categorises expenses, tracks budgets, and generates insights using Python, FastAPI, Clean Architecture, DTO/Mapper/Repository patterns, OCR with GPT-4o, SQLAlchemy, AWS Lambda, API Gateway, and RDS.

RecallAI - Personal Knowledge LLM Assistant

Self-hosted AI assistant for document ingestion, semantic search, and conversational knowledge recall, designed to run with local LLMs and vector search for cost-efficient inference. Tech: Python, FastAPI, FAISS vector search, local LLMs, Oracle Cloud Free Tier, and REST APIs.

ENTERPRISE PLATFORM EXPERIENCE

- Salary Packaging & Financial Systems - enterprise salary-packaging and financial platforms for Smartgroup Australia.
- Employee Benefits Platforms - BrightChoices, Benefit Access, and FSA Portal at Willis Towers Watson.
- Insurance Automation & Customer Portals - backend automation, reporting, and customer-facing insurance systems at AXA Philippines and OneBeacon.
- Banking & Card Systems - credit-card, debit-card, reconciliation, compliance, rewards, and financial-processing platforms for Equitable Bank, Robinsons Bank, and BDO.
- HR & Healthcare Management Systems - HRIS, clinic management, billing, patient-data, and healthcare workflow systems.

EDUCATION

Master of Software Engineering (Level 9)

Yoobee College of Creative Innovation, New Zealand | Completed Aug 2026 | 180 NZQA credits

Industry-Based Capstone Research Project: MemoLink - Context-Aware AI Companion for Knowledge Capture, Contextual Retrieval, and Task Support in Study and Work Settings (A+).

Bachelor of Science in Computer Science

Filamer Christian University, Philippines | 2004 - 2008

PROFESSIONAL INTERESTS

Software architecture, cloud engineering, AI systems, knowledge management, secure full-stack platforms, database modernisation and migration, financial systems, healthcare systems, technical leadership, and ethical technology development.

References: Available upon request.